

Sagargouda Patil

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Bangalore, Karnataka

PROFESSIONAL SUMMARY

Electronics & Communication Engineering graduate with hands-on training and internship experience in **RTL Design, FPGA development, Verilog, VHDL, and digital verification**. Trained across key VLSI domains with primary focus on RTL design and digital systems. Experienced in developing synthesizable RTL, testbenches, FPGA/CPLD implementations, UART-based communication, and hardware integration using Xilinx tools. Familiar with ASIC design flow, synthesis, timing concepts, and industry-standard VLSI EDA tools.

TECHNICAL SKILLS

HDL	Verilog, SystemVerilog, VHDL
RTL Design	Combinational/Sequential Logic, FSM, Counters, Registers, Multiplexers, Datapath & Control Logic
FPGA	Xilinx FPGA/CPLD, FPGA Implementation, JTAG, Simulation
Protocols	UART, SPI, AMBA Fundamentals
VLSI	ASIC Design Flow, RTL-to-Gate Flow, Synthesis, Timing Fundamentals, Physical Design Fundamentals
EDA Tools	Xilinx Tools, C, Quartus Prime, Questa Sim
Programming	C, Python
Tools/OS	Linux, Windows

EXPERIENCE

Capgemini *VLSI Training*

RTL Design Engineer Trainee

- Underwent hands-on training across key **VLSI domains**, with primary focus on RTL Design and Digital Design.
- Designed and implemented synthesizable RTL modules using **Verilog/SystemVerilog**. Developed combinational and sequential logic including **FSMs, counters, registers, multiplexers, and control/datapath logic**.
- Gained exposure to the **ASIC design flow**, including RTL design, simulation, synthesis, timing concepts, and physical design.
- Worked with **Cadence Genus and Innovus** as part of VLSI design-flow training.

Defence Research and Development Organisation (DRDO)

Oct 2023 – Nov 2023

RTL Verification Intern

- Worked on **FPGA and CPLD simulation and implementation** using Xilinx tools.
- Analyzed **VHDL RTL code** and developed Verilog testbenches for functional verification.
- Tested airborne encoder-decoder systems using **UART communication**.
- Performed functional testing of signal transmission and reception under different operating conditions.
- Used **JTAG-based FPGA programming** for hardware integration and testing.

PROJECTS

FPGA-Based Image Histogram Processing *Verilog — FPGA*

- Designed an RTL-based image histogram processing system using **Verilog**.
- Implemented memory-based pixel processing using **\$readmemh** for image data initialization.
- Developed histogram collection logic using pixel values as memory addresses for read, increment, and write-back operations.
- Created a Verilog testbench and analyzed simulation waveforms to verify memory operations and histogram generation.

UART Communication System

Verilog — RTL

- Designed RTL modules for UART transmitter and receiver using **Verilog**.
- Implemented serial data transmission and reception with timing and control logic.
- Developed a testbench to verify UART communication and analyzed waveforms for functional debugging.

EDUCATION

New Horizon College of Engineering, Bangalore **2021 – 2025**

Bachelor of Engineering in Electronics and Communication Engineering **CGPA: 9.56/10**

KLE Prerana PU Science College, Hubli **2019 – 2021**

Pre-University Course (PCMB) **95.6%**

Sri Rajeshwari Vidyaniketan, Hulkoti **2018 – 2019**

CBSE Class X **93%**

TRAINING & LEADERSHIP

- Corporal – NCC** — **Secretary – Alumni Club**
- Fest Lead – Sargam 2024** (National-Level College Fest) and **Team Lead – Aqua Rush, Quantum-X 2024**.